



Deriving and validating emotional dimensions from textual data

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Highlights

- New methodology for deriving orthogonal emotional dimensions.
- Expectancy tension dimension is introduced as a new emotional dimension.
- Pleasure and arousal are confirmed as orthogonal dimensions from previous research.
- Evidence is found towards confirming “novelty” dimension from previous research.

Abstract

This paper proposes and analyzes a methodology for extracting the underlying emotional dimensions connected to different textual data, including social-media posts and online reviews. Our experiments result in a coherent conclusion across all 16 studied datasets. In particular, the found orthogonal emotional dimensions are a combination of *valence* (positive–negative

sentiment), *activation arousal* (arousal–dominance), and *expectancy tension* (the intensity of the expectations concerning the future). We confirm the existence of both *valence* and *arousal* as core dimensions. On the other hand, *dominance* appears as an attribute connected to the variability of both valence and activation arousal dimensions. We also find some evidence for the existence of an “*unpredictability/novelty*” dimension discussed in recent academic work. Our key empirical contribution is that an additional orthogonal emotional dimension should be defined and named “*expectancy tension*” in that it captures the variability linked to the intensity of expectations regarding the future. Finally, our work contributes to the social computing literature by suggesting a novel methodology to derive emotional spaces from multiple textual data through eigenvector analyses.

Introduction

The explosion of social connections and communication channels enabled by the Internet and, more specifically, social networks have created a huge opportunity for (social) scientists interested in exploring human interactions, behaviors, and emotions. Within this context, we leverage textual data from social networks and online reviews to analyze dimensional models of human emotion, which are oftentimes tied to emotional states.

Mehrabian and Russell popularized the idea of emotional dimensional models after introducing the seminal Pleasure–Arousal– Dominance (PAD) model, also known as the Valence–Arousal– Dominance (VAD) model (Bakker et al., 2014). These models create a theoretical dimensional space where all emotions can be mapped to. In other words, emotional dimensional models represent a useful space for emotional mapping. They operate under the assumption that all emotions are not discrete, but instead, they represent transitory neuropsychological states. This, in turn, enables a better understanding of human emotional states and behavior since only a reduced number of representative dimensions needs to be analyzed, as opposed to all potential emotions.

In this paper, we investigate the most common and stable emotional dimensions that can explain the highest amount of variability in different human emotions. To do so, we propose a novel methodology that relies on observational, rather than experimental data.

There is a myriad of different emotions, both basic and those derived from other emotions. The motivating force behind this work comes from the idea that if we can objectively analyze and understand the emotional dimensional space, we might then be able to focus only on a few key dimensions instead of all different individual emotions. Since emotions underpin human behavior, those key dimensions can then be used in future studies by researchers trying to map emotional states onto actions and behaviors.

Our main goals are to propose and validate a methodology for discovering the true orthogonal space where emotions can be mapped to. In particular, unlike previous work, our proposed methodology derives emotional dimensions from empirical, textual data. Having such a well-defined

methodology allows us to confirm previous academic findings and or detect new emotional dimensions stable across many datasets.

Having defined our motivation and objectives, the overarching research questions guiding our work are:

- *RQ1*: How can one methodically validate findings from previous emotional dimensional models based on textual data?
- *RQ2*: What are the orthogonal emotional dimensions that are stable across several textual datasets?

The above research questions indicate that we are leaning on previous research on basic emotions and emotional dimensions and looking at the orthogonality of those dimensions. Technically, we are trying to find core emotional dimensions that are orthogonal. This endeavor leads to two major contributions. First, our results confirm some major conclusions derived by previous research in that we observe the emergence of previously identified emotional dimensions. Second, our methodology is able to detect new relevant dimensions that expand the previous academic findings and better explain basic emotions. For example, we find that a three-dimensional orthogonal model explains the variability of emotional states better than other models do. We call these dimensions *valence* (i.e., positive–negative sentiment), *activation arousal* (i.e., general arousal state), and *expectancy tension* (i.e., the intensity related to scenarios involving expectancy towards the future). The first two dimensions are also commonly found in previous work, but the newly introduced third dimension captures the variability of basic emotions related to expectancy, e.g., anticipation and surprise.

To derive the above results, we assume that short texts, such as social media posts and online reviews, are valid proxies that capture users' current moods and emotional states. That is, they provide a snapshot of human emotional states. We argue that such an assumption is valid due to the general length of the studied texts and straightforward language. As such, one should be able to successfully detect dimensional models and their components from the analyzed textual data.

Besides this introductory section, the rest of this paper is organized as follows. Section2 elaborates on previous research discoveries related to emotions and emotional dimensions. Section3 explains the studied datasets and our novel methodology to derive emotional dimensions from observational data. Sections4 Experimental results, 5 Discussion introduce and elaborate on our main findings. We finally conclude and provide an outlook on future work in Section6.

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Section snippets

Research background & related work

Emotions, or emotional states, have been identified as discrete psychological states by several researchers, which led to the now established *discrete emotion theory*. The discrete emotion theory identifies emotions from their expressive features based on analysis at the neurological and expression level (Colombetti, 2009). Several models of emotion fall into this category. Notably, the work by Plutchik (1980) identified eight basic emotions, namely *acceptance*, *anger*, *anticipation*, *disgust*, *joy*, ...

Emotional dimensions & observational data

As we mentioned before, we focus on implementing a novel methodology to discover stable dimensions of emotions across many different textual datasets from different sources. It is then crucial to carefully elaborate on our choices regarding datasets, which we do in Section 3.1. In Section 3.2, we further elaborate on our choice regarding using PCA for analysis, specifically, robust sparse PCA. Moreover, we explain our novel methodology in Section 3.3.

In short, we employ two different techniques ...

Experimental results

In what follows, we present some key findings after applying our proposed methodology to the datasets we described in Section 3.1. We argue that our results are generalizable because we apply our methodology to multiple datasets with different topics that induce various emotional states. Finally, throughout this section, we contrast the results found by our methodology with previous results. In particular, we are generally guided by three key academic works:

- The PAD dimensional model by Russell ...

...

Discussion

In this section, we combine the results from our individual analysis in the previous section to provide a unified explanation of our findings. We start by noting that both analyses in our proposed methodology, hierarchical clustering and global PCA analysis, resulted in a consistent conclusion, namely the existence of three emotional dimensions. In other words, three emotional dimensions are consistent in our analysis across different textual datasets and statistical techniques. In particular, ...

Conclusion

While previous academic research about dimensional models of emotion has predominately relied upon surveys and relatively small sample sizes, we propose in this paper a new methodology to test and derive emotional dimensions from textual data. In particular, we analyzed datasets having hundreds of thousands of observations. From product reviews, general elections, racial tensions, all the way to global pandemics and natural disasters, these datasets were intentionally selected since they ...

Supplementary materials

For additional information, we refer the interested reader to sociallab.fer.hr ↗. ...

CRedit authorship contribution statement

Demijan Grgić: Conceptualization, Methodology, Investigation, Formal analysis, Writing – original draft. **Vedran Podobnik:** Supervision. **Arthur Carvalho:** Supervision, Validation, Writing – review & editing. ...

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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